

UK IT Innovation

Gold

Taos Operating System 1.27
from Taos Systems

Silver: White Cross WS9010 from
White Cross Systems

Bronze: ARM 700 family Risc
micro-processors from
Advanced Risc Machines

It's not often that a new software product gets rave reviews in publications as diverse as entertainment specialist title *Edge Magazine* and techie journal *Risc*. But Taos Systems, developer of a revolutionary operating system called Taos, is used to it. Not being easy to pigeonhole is part of the charm of being an innovative company.

But it can also be one of the drawbacks, especially when it comes to raising funds. In the early days, Taos' inventor Chris Himsley financed the new operating system by developing computer and video games. When the original funding collapsed, Taos' current chairman Francis Churig stepped in.

'A lot of innovation is left as innovation in this country,' Churig says. 'I've been round the world with this product, and seen the different reactions in different countries. There is definitely a reluctance for people in the UK to believe that a British product can be any good, and that it can be commercially successful. Venture capitalists take no risks in this country.'

Churig eventually secured funding from Japanese investors, with whom Taos Systems has set up a joint global distribution company called TK5. 'Our emphasis will be on the Far East, not Europe, in the beginning,' Churig says. 'Taos is a product for all seasons, but it is particularly appropriate for the multimedia market, which happens to be dominated by Far Eastern and Japanese companies.'

Taos is innovative because it is wholly portable. According to the company, it is a compact operating system kernel for single and parallel processor systems. Applications for Taos need to be written only once, after which they can run on any hardware platform without being re-compiled.

The implications for games and other software manufacturers are huge, Churig says, because only the format needs to be changed when they release products for different hardware environments.

Using Taos, developers can write for familiar environments such as the PC, yet their applications can also run on Advanced



Taos' Francis Churig (right): 'There is definitely a reluctance for people in the UK to believe that a British product can be any good.'

Risc Machines (ARM) and Irmes processors. Developers don't need to know how many parallel processors will be required by the application, since a built-in load-balancing feature enables applications to exploit additional power as it is added.

One of Taos' target markets will be the flourishing television set-top box market, which will be driven by a move towards interactive television and cable services on the information superhighway.

'The advantage of Taos for this market is its microkernel architecture,' says Churig. 'Instead of taking up 8Mb of Ram in a set-top box, it uses only 1Mb. Manufacturers appreciate this, because they can bring down the total cost of their boxes.'

For the same reason, Taos is being used by a number of manufacturers as an embedded operating system for personal digital assistants (PDAs), arcade games, virtual reality machines and robotic systems. Around 40 organisations are using Taos so far, including Goleconda Games and various Japanese multimedia companies.

Taos Systems is conspicuously filling niches not yet targeted by other, larger operating system vendors, particularly Microsoft. 'We're trying to latch onto the

markets where Microsoft is not yet fully involved,' Churig says. 'We are going straight to the top of global industries, such as entertainment, multimedia and communications. We're not going to let this opportunity slip.'

The challenge for Taos Systems, as it enters a new phase of development for Windows and Posix application programming interfaces (APIs), will be to apply its technology in the right way.

'We have a product which could be seen as highly esoteric and difficult to understand,' says Churig. 'What we need to aim for are products which are as close to the complexity level of the Videopass video recording system as possible. There's no point at all in having the functionality if people can't access it easily.'

Churig's final word of advice to other would-be UK innovators is to be persistent. 'Don't give up,' he says. 'And don't sell out too soon. Someone, somewhere will eventually believe in you. One of the first things we intend to do when we become a highly successful, cash-rich software company is to help UK innovation. There are so many potential successes which never see the light of day.'